# Finite Certificates for Stern Roots and a Bilinear Hankel Identity

Complete arguments, auxiliary classifications, and a five-target audit

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Research round T86–T90

**Abstract**

This manuscript records a five-target research round, separating exact prior-question applications from auxiliary results and unsuccessful attempts. A parity-separated quadratic envelope, combined with finite polynomial positivity certificates, proves the entire three-family assertion in Ulas's Conjecture 4.3(1): the known zero $-1$ is the unique real zero and is simple. For Cigler's Conjecture 4.1, we reduce boundary-deformed path Hankel determinants to a reciprocal Cauchy determinant. Its spectral root equation identifies three classical Miwa directions with the integer shifts $1, -k, k+1$, yielding the original bilinear recurrence for every parameter and every permitted determinant size. An exact classification of Laplacian peak transfer on uniform two-level trees is included as an auxiliary result; the underlying existence question has a prior public solution. The least-dimensional Ehrhart counterexample question and the all-odd-period PT existence conjecture remain unresolved here. No historically unprecedented foundational principle, exhaustive priority determination, or five independent open-problem solutions are claimed.

## 1 The five targets and what is actually established

The labels T86–T90 identify research tracks, not a promise of five successful theorems. We require an unchanged, previously stated target before counting an application as a prior-question resolution. Supporting lemmas, already known solutions, special cases, and unsuccessful constructions are not added to that count.

| Track | Original target | Outcome in this manuscript |
|---|---|---|
| T86 | Clarke Question 6.8: least dimension of an ineffective polynomial equivariant numerator. | Unresolved. A concrete candidate is effective; it is not a counterexample. |
| T87 | Faust et al. Conjecture 2: nonzero PT-symmetric zero-isospectral potentials for every period greater than three. | Unresolved. A complete obstruction eliminates one sparse construction for all odd periods at least seven. |
| T88 | Ulas Conjecture 4.3(1): exactly one real root in each of three original Stern families. | Complete proof for all three families and all $n \geq 2$, including simplicity and positivity of the quotient. |
| T89 | Coutinho–Guo–Schmeits Open Problem 7.1: infinitely many non-star trees with Laplacian peak transfer. | Excluded from new-resolution counting because of prior solution. A full classification inside a two-parameter tree family is proved. |
| T90 | Cigler Conjecture 4.1: an all-column bilinear Hankel recurrence. | Complete proof of the original recurrence, including exceptional parameters and vanishing determinants. |

The principal mathematical claims are the unchanged Ulas clause and Cigler recurrence. Neither result is a proof of every conjecture in its source. The methods use classical Stern identities, concavity, Bernstein positivity, Christoffel determinants, and the Hirota–Miwa identity. The additional content proposed here consists of the finite certificates for the specific infinite Stern families and the exact reciprocal-root reduction of the Hankel problem. These applications do not establish a historically new foundational principle. A bounded literature search cannot prove that no earlier identical solution exists.

All statements presented as theorems below have complete unconditional arguments. This is an internally checked research draft, not an externally refereed paper or a proof-assistant development. The user's full set of historical success conditions is therefore not certified.

## 2 Finite envelopes and the real-root conjecture for three Stern families

Let $B_0(t) = 0$, $B_1(t) = 1$, $B_{2m}(t) = tB_m(t)$, and $B_{2m+1}(t) = B_m(t) + B_{m+1}(t)$. The original target is Conjecture 4.3(1) of Ulas [1], for the unchanged indices

$$\begin{aligned} s_{0,n} &= 2^{2n+4} - 9 \cdot 2^{n+1} - 1, \\ s_{2,n} &= 2^{2n+8} - 9 \cdot 2^{n+4} - 13, \\ s_{3,n} &= 2^{2n+8} - 51 \cdot 2^{n+2} - 1. \end{aligned}$$

That numbered assertion says that each $B_{s_{i,n}}$ has exactly one real root for $i \in \{0, 2, 3\}$ and every $n \geq 2$. We prove the stronger strict positivity of the quotient by its known factor $t + 1$. This is different from the eventual-monotonicity counterexample in T81. It does not assert the irreducibility asked for in Conjecture 4.3(2), or settle the other clauses of Conjecture 4.3.

### 2.1 The auxiliary finite-certificate problem

The auxiliary problem is to certify the sign of an infinite family

$$F_n(x) = a(x) + b(x)(-x)^n + c(x)x^{2n}, \qquad n \geq 2,$$

without bounding $n$. Splitting by parity turns the moving power into one real variable in a known interval. The following observation gives a finite reduction when the resulting quadratic is concave.

**Lemma 2.1** (Parity-separated quadratic envelopes)**.** *Fix $x > 0$ and put $g_e(v) = a(x) + (-1)^e b(x)v + c(x)v^2$ for $e \in \{2, 3\}$. For $n \geq 2$ of the same parity as $e$, we have $F_n(x) = g_e(x^n)$. If $0 < x < 1$ and $c(x) \leq 0$, then*

$$F_n(x) \geq \min\{a(x), F_e(x)\}.$$

*If $x > 1$, $c(x) < 0$, and $g_e'(x^e) < 0$, then*

$$F_n(x) \leq F_e(x).$$

*Proof.* In the first case, $0 < x^n \leq x^e$ and a concave quadratic on $[0, x^e]$ takes its minimum at an endpoint. In the second case, $x^n \geq x^e$ and $g_e'(v) = (-1)^e b(x) + 2c(x)v$ is strictly decreasing and already negative at $x^e$. □

The remaining finite positivity assertions below use the elementary Bernstein basis

$$\mathcal{B}_{j,N}(u) = \binom{N}{j} u^j (1-u)^{N-j}, \qquad 0 \le j \le N.$$

If $p(ru) = D^{-1} \sum_{j=0}^{N} v_j \mathcal{B}_{j,N}(u)$ with $D > 0$, $v_j \ge 0$, and $v_0, v_N > 0$, then $p(x) > 0$ on $[0, r]$. Indeed all basis functions are positive on $(0, 1)$, and the endpoint values are $v_0/D$ and $v_N/D$. This proof uses only finite polynomial identities; no numerical root isolation is assumed.

## 2.2 Exact family formulas and finite positivity data

Put $A(x) = 2x^3 + 2x^2 - 3x + 1$ and write

$$F_{i,n}(x) := (1+x)^2 B_{s_{i,n}}(-x) = a_i(x) + b_i(x)(-x)^n + c_i(x)x^{2n}. \tag{S1}$$

The coefficient polynomials are

$$\begin{aligned}
&a_0 = A, \qquad b_0 = -2(1-x)^2, \qquad c_0 = -x(x^2 - x + 2),\\
&a_2 = (2x^2 - 2x + 1)A, \qquad c_2 = x^2(2-3x)(x^2 - x + 2),\\
&b_2 = 2x(1-x)^2(x^3 + x^2 - 3x + 2),\\
&a_3 = 2x^5 - 4x^3 + 6x^2 - 3x + 1,\\
&b_3 = 2x(1-x)(3x^2 - 2x + 1), \qquad c_3 = -x^3(5x^2 - 5x + 2).
\end{aligned}$$

Here and below the argument $x$ on the right is suppressed when unambiguous. For completeness, these identities follow from the Stern addition formula

$$B_{m2^a - r} = B_{2^a - r} B_m + B_r B_{m-1} \qquad (0 \le r \le 2^a),$$

and its plus version. A convenient way to evaluate the complementary indices is, for $2^b \ge r$ and $a \ge b$,

$$B_{2^a - r}(t) = \frac{t^{a-b}\{B_r(t) + (t-1)B_{2^b - r}(t)\} - B_r(t)}{t-1}. \tag{S2}$$

The plus formula with $2^a - r = 2^b(2^{a-b} - 1) + (2^b - r)$ proves (S2) directly; all displayed quotients are polynomials. For $i = 0, 2, 3$, respectively, apply the minus formula to

$$s_{0,n} = 2^{n+1}(2^{n+3} - 9) - 1, \quad s_{2,n} = 2^{n+4}(2^{n+4} - 9) - 13, \quad s_{3,n} = 2^{n+2}(2^{n+6} - 51) - 1.$$

Then use (S2) with $r = 9, 10, 13, 51, 52$ and $b = 4, 4, 4, 6, 6$. The small polynomials needed are

$$B_9 = (t+1)^2,\ B_{10} = t + 2t^2,\ B_{13} = 1 + 2t + 2t^2, \quad B_{51} = 1 + 3t + 5t^2 + 3t^3, \quad B_{52} = t^2 + 2t^3 + 2t^4.$$

Substitution and multiplication give (S1) for every $n \ge 2$, rather than merely a polynomial interpolation in $n$.

For $e = 2, 3$ define the following six base quotients by the identities

$$F_{i,e}(x) = (1-x)(1+x)^2 q_{i,e}(x). \tag{S3}$$

Explicitly,

$$\begin{aligned}
q_{0,2} &= x^4 - 2x^3 + 5x^2 - 4x + 1,\\
q_{0,3} &= x^6 - 2x^5 + 5x^4 - 6x^3 + 7x^2 - 4x + 1,\\
q_{2,2} &= 3x^6 - 10x^5 + 23x^4 - 26x^3 + 17x^2 - 6x + 1,\\
q_{2,3} &= 3x^8 - 8x^7 + 21x^6 - 32x^5 + 37x^4 - 30x^3 + 17x^2 - 6x + 1,\\
q_{3,2} &= 5x^6 - 10x^5 + 17x^4 - 16x^3 + 11x^2 - 4x + 1,\\
q_{3,3} &= 5x^8 - 10x^7 + 17x^6 - 22x^5 + 23x^4 - 18x^3 + 11x^2 - 4x + 1.
\end{aligned}$$

All six are strictly positive on $[0,\infty)$. To give finite exact certificates, first note

$$q_{0,2}(x) = (x^2 - x)^2 + (2x - 1)^2 > 0 \qquad (x \in \mathbb{R}).$$

For the other five, the following table records $(N, D, (v_j)_{j=0}^{N})$ in the identity

$$q(x) = D^{-1} \sum_j v_j \mathcal{B}_{j,N}(x).$$

| Polynomial | $N$ | $D$ | $(v_0, \ldots, v_N)$ |
|---|---|---|---|
| $q_{0,3}$ | 6 | 30 | $(30, 10, 4, 3, 8, 20, 60)$ |
| $q_{2,2}$ | 6 | 30 | $(30, 0, 4, 3, 4, 10, 60)$ |
| $q_{2,3}$ | 8 | 140 | $(140, 35, 15, 5, 4, 5, 20, 70, 420)$ |
| $q_{3,2}$ | 6 | 15 | $(15, 5, 6, 6, 10, 15, 60)$ |
| $q_{3,3}$ | 8 | 140 | $(140, 70, 55, 50, 56, 65, 90, 105, 420)$ |

These identities prove positivity on $[0,1]$. On $[1,\infty)$, the coefficient vectors of $q(1+y)$, in ascending powers of $y$, are

| Polynomial | Coefficients of $q(1+y)$ |
|---|---|
| $q_{0,2}$ | $(1, 4, 5, 2, 1)$ |
| $q_{0,3}$ | $(2, 8, 14, 14, 10, 4, 1)$ |
| $q_{2,2}$ | $(2, 10, 22, 26, 18, 8, 3)$ |
| $q_{2,3}$ | $(3, 20, 60, 106, 122, 94, 49, 16, 3)$ |
| $q_{3,2}$ | $(4, 18, 40, 52, 42, 20, 5)$ |
| $q_{3,3}$ | $(3, 18, 60, 124, 168, 150, 87, 30, 5)$ |

Their positive coefficients prove the remaining assertion.

We also need $a_i > 0$ on $[0,1]$. For $a_0 = A$ and $a_3$, their Bernstein certificates have degree five, denominator five, and vectors

$$A: \ (5, 2, 0, 0, 3, 10), \qquad a_3: \ (5, 2, 2, 3, 3, 10).$$

The factor $2x^2 - 2x + 1 = 2(x - 1/2)^2 + 1/2$ proves the assertion for $a_2$. Finally put

$$J(x) = a_2(x) - b_2(x)x^3. \tag{S4}$$

The exact degree-nine Bernstein certificate for $J(2u/3)$ has denominator 413343 and vector

$$(413343, 260253, 158193, 95499, 57915,$$
$$36369, 26973, 28623, 47159, 97125).$$

Thus $J(x) > 0$ for $0 \le x \le 2/3$. Every entry in these tables is rational or integral; their defining identities can be verified by expansion and form part of the proof.

### 2.3 The full original real-root assertion

**Theorem 2.2** (Ulas Conjecture 4.3(1), with a positive quotient)**.** *For every $n \ge 2$ and $i \in \{0, 2, 3\}$,*

$$\frac{B_{s_{i,n}}(t)}{t+1} > 0 \qquad (t \in \mathbb{R}),$$

*where the quotient is the polynomial obtained by division by $t + 1$. Consequently $-1$ is the unique real zero of $B_{s_{i,n}}$, and that zero is simple. This proves the whole assertion of [1, Conjecture 4.3(1)].*

*Proof.* For $t \geq 0$, the Stern recurrence gives nonnegative coefficients and constant coefficient one at these odd indices, so the asserted quotient is positive. For $t = -x < 0$, it suffices to prove $F_{i,n}(x)/(1-x) > 0$, treating $x = 1$ by its removable value.

Suppose first $0 < x < 1$. For $i = 0, 3$ we have $c_i(x) < 0$. For each parity take $e = 2$ or $e = 3$ in Lemma 2.1. Both endpoints are positive: $a_i(x) > 0$ by the certificates, and $F_{i,e}(x) = (1-x)(1+x)^2 q_{i,e}(x) > 0$. Thus $F_{i,n}(x) > 0$. For $i = 2$ the same argument applies on $2/3 \leq x < 1$, because $c_2 \leq 0$ there. On $0 < x \leq 2/3$, we have $c_2 \geq 0$ and $b_2 \geq 0$, since $x^3 + x^2 - 3x + 2 \geq 2 - 3x \geq 0$. If $n$ is even, all three terms in (S1) are nonnegative and $a_2 > 0$. If $n$ is odd, then $n \geq 3$, $x^n \leq x^3$, and

$$F_{2,n}(x) = a_2 - b_2 x^n + c_2 x^{2n} \geq a_2 - b_2 x^3 = J(x) > 0.$$

This proves the required sign for every family on $(0, 1)$.

Now suppose $x > 1$. All three $c_i$ are negative. We check the second condition in Lemma 2.1 at $e = 2, 3$. For $i = 0$, $b_0 < 0$, so the even derivative is negative. The odd one is

$$-b_0 + 2c_0 x^3 = 2(x-1)^2 - 2x^4(x^2 - x + 2) < 0.$$

Here $x^2 - x + 2 \geq 2$ and $x^4 > (x-1)^2$. For $i = 2$, $b_2 > 0$ on $x > 1$, since $x^3 + x^2 - 3x + 2 = x(x^2 - 1) + (x-1)^2 + 1 > 0$. Thus the odd derivative is negative. The even derivative is $-2xH_2(x)$, where

$$H_2 = x^3(3x - 2)(x^2 - x + 2) - (x-1)^2(x^3 + x^2 - 3x + 2).$$

For $i = 3$, $b_3 < 0$, so the even derivative is negative; the odd one is $-2xH_3(x)$, where

$$H_3 = x^5(5x^2 - 5x + 2) - (x-1)(3x^2 - 2x + 1).$$

The positivity needed here follows from the exact expansions

$$H_2(1+y) = 2 + 13y + 30y^2 + 36y^3 + 24y^4 + 12y^5 + 3y^6,$$
$$H_3(1+y) = 2 + 13y + 46y^2 + 92y^3 + 110y^4 + 77y^5 + 30y^6 + 5y^7.$$

Thus for the matching parity,

$$F_{i,n}(x) \leq F_{i,e}(x) = (1-x)(1+x)^2 q_{i,e}(x) < 0.$$

This proves positivity of the quotient also when $x > 1$.

It remains to verify division and strict positivity at $x = 1$. Formula (S1) gives $F_{i,n}(1) = 0$, proving divisibility by $t + 1$. Differentiation of that same identity gives the removable values

$$\left.\frac{B_{s_{0,n}}(t)}{t+1}\right|_{t=-1} = n - 1, \qquad \left.\frac{B_{s_{2,n}}(t)}{t+1}\right|_{t=-1} = n, \qquad \left.\frac{B_{s_{3,n}}(t)}{t+1}\right|_{t=-1} = n + 1 + (-1)^n.$$

They are strictly positive for $n \geq 2$. The quotient therefore never vanishes on $\mathbb{R}$, and the root $-1$ is simple. □

**Attribution and scope.** The underlying Stern addition identities are prior. Concavity, parity splitting, and Bernstein positivity are classical tools. The proposed additional content is their finite-envelope application to all three original families, including the convex subinterval of the second family and the simple-root boundary values. No historically unprecedented general principle is inferred from this application. A bounded source search did not locate an earlier proof of this precise three-family statement; this is not a guarantee of exhaustive priority.

### 2.4 The fourth family: an exact classification and the multiplicity issue

For completeness consider the remaining original family

$$s_{1,n} = 2^{2n+5} - 9 \cdot 2^{n+2} - 5, \qquad n \geq 1.$$

Conjecture 4.3(3) of Ulas [1] says that $B_{s_{1,n}}$ has exactly three real roots for $n \geq 3$. The exact classification below distinguishes distinct roots from roots counted with multiplicity. In particular, $n = 3$ has only one distinct real root, of multiplicity three. We retain that endpoint rather than quietly replacing the original range by $n \geq 4$.

**Lemma 2.3** (A uniform factorization for the fourth family)**.** *Put*

$$M_n(t) = (t+2)t^n - 2t - 1, \qquad N_n(t) = (t^2+t+2)t^n + 2t^3 - 2t^2 - 3t - 1.$$

*For every $n \geq 1$,*

$$B_{s_{1,n}}(t) = \frac{M_n(t)N_n(t)}{(t-1)^2}. \tag{S5}$$

*Both quotients $M_n/(t-1)$ and $N_n/(t-1)$ are polynomials.*

*Proof.* Write $s_{1,n} = 2^{n+2}(2^{n+3} - 9) - 5$. The addition identity gives

$$B_{s_{1,n}} = B_{2^{n+2}-5}B_{2^{n+3}-9} + B_5 B_{2^{n+3}-10}.$$

The complement identity (S2) yields

$$B_{2^{n+2}-5} = \frac{M_n}{t-1}, \qquad B_{2^{n+3}-10} = \frac{tM_n}{t-1}, \qquad B_{2^{n+3}-9} = \frac{t^n(t^2+t+2) - (t+1)^2}{t-1}.$$

Since $B_5 = 1 + 2t$, their substitution proves (S5). This is a recurrence derivation, independent of the displayed parity-dependent factors in the source. □

**Theorem 2.4** (Complete real-root classification of the fourth family)**.** *For $n \geq 4$, $B_{s_{1,n}}$ has exactly three real roots, all simple:*

$$-r_n^{-1}, \qquad -1, \qquad -r_n, \qquad 0 < r_n < 1.$$

*If $n$ is even, then $0 < r_n < 1/2$; if $n$ is odd, then $1/2 < r_n < 1$. The exceptional initial cases are*

$$\begin{aligned} B_{s_{1,1}} &= (t+1)(3t^2+2t+1), \\ B_{s_{1,2}} &= (t+1)(t^2+3t+1)^2, \\ B_{s_{1,3}} &= (t+1)^3(t^4+2t^3+6t^2+4t+1). \end{aligned}$$

*Thus the numbers of distinct real roots for $n = 1, 2, 3$ are respectively $1, 3, 1$, with multiplicities respectively $(1)$, $(2, 1, 2)$, and $(3)$. For every $n \geq 3$ the number of real roots counted with multiplicity is exactly three.*

*Proof.* The three initial factorizations follow from Lemma 2.3. Their quadratic factor at $n = 1$ has negative discriminant. The quartic factor at $n = 3$ is positive on the real line: at $t = -x$ it is

$$x^4 - 2x^3 + 6x^2 - 4x + 1 = (x^2 - x)^2 + (2x - 1)^2 + x^2 > 0.$$

The strictness follows because these three squares cannot vanish simultaneously. The quadratic at $n = 2$ has two distinct negative roots, neither equal to $-1$.

Suppose first that $n \geq 4$ is even. Then

$$B_{s_{1,n}} = (t+1)\,\frac{M_n}{t-1}\,\frac{N_n}{t^2-1}.$$

We show that the last quotient is positive on the real line. For $t = -x$ with $x > 0$ and $x \neq 1$, it equals

$$G_n(-x) = \frac{(x^2-x+2)x^n - A(x)}{x^2-1}, \qquad A(x) = 2x^3 + 2x^2 - 3x + 1.$$

Since $n \geq 4$, comparison with $x^4$ shows $G_n(-x) \geq G_4(-x)$ on both $0 < x < 1$ and $x > 1$. The base polynomial satisfies

$$\begin{aligned} G_4(-x) &= x^4 - x^3 + 3x^2 - 3x + 1 \\ &= \left(x^2 - \frac{x}{2}\right)^2 + \frac{11}{4}\left(x - \frac{6}{11}\right)^2 + \frac{2}{11} > 0. \end{aligned}$$

At $x = 1$, differentiation of numerator and denominator gives $G_n(-1) = n - 3 > 0$. For $t \geq 0$, positivity also follows from

$$\frac{N_n(t)}{t-1} = B_{2^{n+3}-9}(t) + tB_5(t) > 0,$$

and $t + 1 > 0$.

It remains to count zeros of $M_n/(t-1)$. This is the reciprocal polynomial

$$\frac{M_n(t)}{t-1} = 1 + 3(t + t^2 + \cdots + t^{n-1}) + t^n.$$

At $t = -x$ it equals

$$\frac{1 - 2x - (2-x)x^n}{x+1}.$$

For $0 < x < 1$, its numerator has derivative

$$-2 - x^{n-1}\{2n - (n+1)x\} < 0.$$

It takes values 1 at $x = 0$, a negative value at $x = 1/2$, and $-2$ at $x = 1$. Thus it has exactly one zero $r_n \in (0, 1/2)$, and that zero is simple. Reciprocity gives exactly one further zero at $1/r_n > 1$, also simple; $x = 1$ is not a zero. There are no nonnegative roots because the coefficients are positive. This proves the even case, including simplicity of the factor $t + 1$.

Now suppose $n \geq 5$ is odd. Then

$$B_{s_{1,n}} = (t+1)\,\frac{M_n}{t^2-1}\,\frac{N_n}{t-1}.$$

The last quotient is positive. Indeed, for $t = -x$ with $x > 0$ it is

$$\frac{A(x) + (x^2-x+2)x^n}{x+1} > 0.$$

We already proved $A(x) > 0$ on $[0, 1]$; for $x = 1+y \geq 1$, the identity $A(1+y) = 2+7y+8y^2+2y^3$ proves the rest. At $t \geq 0$ the same Stern-polynomial expression as above proves positivity.

The remaining factor $K_n = M_n/(t^2-1)$ is reciprocal of even degree $n-1$. This follows either by division or from $t^{n+1}M_n(1/t) = -M_n(t)$ and the same anti-reciprocity of $t^2 - 1$. At $t = -x$ it is

$$K_n(-x) = \frac{f_n(x)}{x^2-1}, \qquad f_n(x) = 2x - 1 - (2-x)x^n.$$

On $0 < x < 1$,

$$f_n''(x) = -nx^{n-2}\{2(n-1) - (n+1)x\} < 0.$$

Furthermore $f_n(0) = -1$, $f_n(1) = 0$, $f_n'(0) = 2$, and $f_n'(1) = 3 - n < 0$. The strictly decreasing first derivative therefore has exactly one zero. The function increases to its unique maximum and then decreases to zero at 1; since the maximum is positive, there is exactly one interior zero, on the strictly increasing part. It is simple. As $f_n(1/2) < 0$, that zero $r_n$ lies in $(1/2, 1)$. Reciprocity supplies exactly one further simple zero at $1/r_n$. At the removable point,

$$K_n(-1) = \frac{3-n}{2} < 0,$$

so $-1$ is not a zero of $K_n$. Finally, $K_n(t) > 0$ for $t \geq 0$: $M_n/(t-1)$ has positive coefficients, and division by $t + 1 > 0$ preserves its positive values. This proves the odd case and completes the full classification. □

**Exact relationship with the original statement.** Theorem 2.4 proves the stated three-root assertion on its full range $n \geq 3$ when roots are counted with multiplicity. Under a distinct-root interpretation, the endpoint $n = 3$ refutes it, and the theorem gives the exact corrected classification. The source does not explicitly specify multiplicity in Conjecture 4.3(3), although Conjecture 4.2 separately defines a count using set cardinality. For that reason this is not counted as another unqualified resolution of an unambiguous original conjecture.

The inspected arXiv v1, page 16, also has a coefficient discrepancy in its odd-index factorization: in its first factor, the coefficient of $t(t^{2(m+1)} - 1)/(t-1)$ must be 4, not 1. For example $m = 1$ already distinguishes the two formulas. Identity (S5) provides the corrected all-$n$ expression directly from the recurrence; no conclusion depends on the printed factorization. This observation is about the inspected preprint, not an uninspected journal revision.

# 3 A boundary spectral constraint and Cigler's bilinear Hankel recurrence

The target is Conjecture 4.1 of Cigler's *Some remarks and conjectures about Hankel determinants* [2]. Cigler also posed the identical recurrence in his MathOverflow question of 18 October 2023 [3]. Let $a_{n,j} = a_{n,j}(x, c)$ be defined by

$$a_{0,j} = \delta_{0j}, \qquad a_{n,-1} = 0, \qquad s_0 = c + x, \quad s_j = c \ (j > 0),$$
$$a_{n,j} = a_{n-1,j-1} + s_j a_{n-1,j} + a_{n-1,j+1}.$$

Write

$$D_{k,n}(x, c) = \det(a_{i+j,k}(x, c))_{0 \leq i,j < n}, \qquad D_{k,0} = 1.$$

The original conjecture is a three-term bilinear identity for every $k \geq 1$ and $n \geq 2k + 2$. The auxiliary problem developed below asks which spectral constraints make a Cauchy determinant obey that precise one-dimensional identity. The bridge is a finite boundary condition for orthogonal polynomials, which selects three Miwa shifts with the required integer displacements. The Cauchy determinant and Hirota–Miwa identity are classical ingredients [5]; the proof below is self-contained and does not assert that these general principles are new.

**Theorem 3.1** (Cigler's boundary Hankel recurrence)**.** *For every integer $k \geq 0$, every integer $n \geq 2k + 2$, and all complex parameters $x, c$,*

$$D_{k,n}D_{k,n-2k-2} - x^2 D_{k,n-1}D_{k,n-2k-1} + (x^2 - 1)D_{k,n-k-1}^2 = 0. \tag{1}$$

*All determinants in this formula have the same parameters $(x, c)$. In particular, the unchanged recurrence in Cigler's Conjecture 4.1 holds, including parameter specializations and indices where determinants vanish.*

### 3.1 The auxiliary determinant problem

We first derive the precise determinant identity needed for the transfer. Let $p_i, q_i, w_i$ be parameters, $1 \le i \le s$, with $p_i \neq q_j$. Set

$$C_{ij} = \frac{(p_i - q_i)w_i}{p_i - q_j}, \qquad T = \det(I + C).$$

For a direction $a$, write $T_a$ for the result of replacing each $w_i$ by $w_i(a - p_i)/(a - q_i)$. Multiple subscripts mean that these replacements are combined.

**Lemma 3.2** (Three-direction Cauchy identity)**.** *For distinct directions $a, b, c$, avoiding all $q_i$,*

$$(a - b)T_{ab}T_c + (b - c)T_{bc}T_a + (c - a)T_{ac}T_b = 0. \tag{2}$$

*Proof.* Put $v_i = (p_i - q_i)w_i$, $M = I + C$, and $r_a = (1/(a - q_i))_{i=1}^s$. We work first where $M$ is invertible; the resulting rational identity then extends wherever its expressions are defined. Write $P = \mathrm{diag}(p_i)$ and set

$$F_a = 1 - r_a^{\mathsf{T}} M^{-1} v, \qquad G_a = a - r_a^{\mathsf{T}} M^{-1} P v.$$

Conjugating the matrix defining $T_a$ by $\mathrm{diag}(a - q_i)$ changes it to $M - v r_a^{\mathsf{T}}$, since

$$\frac{a - p_i}{(p_i - q_j)(a - q_j)} = \frac{1}{p_i - q_j} - \frac{1}{a - q_j}.$$

The matrix determinant lemma therefore gives $T_a/T = F_a$. For the two-direction shift the same conjugation, now by $\mathrm{diag}((a - q_i)(b - q_i))$, gives

$$M + P v\, u_{ab}^{\mathsf{T}} + v\, v_{ab}^{\mathsf{T}}, \qquad u_{ab} = \frac{r_b - r_a}{a - b}, \qquad v_{ab} = \frac{b r_a - a r_b}{a - b}.$$

Indeed, subtracting $1/(p_i - q_j)$ from the transformed Cauchy entry leaves $(p_i + q_j - a - b)/((a - q_j)(b - q_j))$ times $v_i$. Applying the rank-two determinant lemma and expanding its $2 \times 2$ determinant yields

$$(a - b)\frac{T_{ab}}{T} = G_a F_b - G_b F_a.$$

After division by $T^2$, the left side of (2) is

$$(G_a F_b - G_b F_a)F_c + (G_b F_c - G_c F_b)F_a + (G_c F_a - G_a F_c)F_b = 0.$$

This proves the identity without any assumption that a shifted determinant is nonzero. □

**Lemma 3.3** (Boundary constraint selecting integer shifts)**.** *Fix $K \ge 2$ and $x \neq 0, \pm 1$. Suppose the nonzero parameters $\lambda_1, \dots, \lambda_s$ satisfy*

$$\lambda_i^K - \lambda_i^{-K} = x(\lambda_i^{K-1} - \lambda_i^{-(K-1)}). \tag{3}$$

*Assume the following determinants and direction shifts are defined. For arbitrary amplitudes $w_i$, let*

$$\tau_m = \det\left(\delta_{ij} + \frac{(\lambda_i - \lambda_i^{-1})w_i \lambda_i^{2m}}{\lambda_i - \lambda_j^{-1}}\right)_{i,j=1}^{s}.$$

*Then, for every integer $m$,*

$$\tau_{m+K}\tau_{m-K} - x^2 \tau_{m+K-1}\tau_{m-K+1} + (x^2 - 1)\tau_m^2 = 0. \tag{4}$$

*Proof.* In Lemma 3.2, use $p_i = \lambda_i$, $q_i = \lambda_i^{-1}$ and the three directions $a = 0$, $b = x$, $c = x^{-1}$. The corresponding amplitude ratios are

$$\frac{0-\lambda_i}{0-\lambda_i^{-1}} = \lambda_i^2, \qquad \frac{x-\lambda_i}{x-\lambda_i^{-1}} = \lambda_i^{-2(K-1)}, \qquad \frac{x^{-1}-\lambda_i}{x^{-1}-\lambda_i^{-1}} = \lambda_i^{2K}.$$

The last two identities follow by rearranging (3) as

$$\lambda_i^{2K-1}(\lambda_i - x) = 1 - x\lambda_i.$$

Thus the three directions increase the determinant index by 1, $-(K-1)$, and $K$, respectively. The three-direction identity becomes

$$-x\,\tau_{r-K+2}\tau_{r+K} + (x - x^{-1})\tau_{r+1}^2 + x^{-1}\tau_{r+K+1}\tau_{r-K+1} = 0.$$

Multiplication by $x$ and the substitution $m = r + 1$ give (4). □

The auxiliary statement allows arbitrary amplitudes, not merely the special coefficients coming from Cigler's moments. Its role is to turn an exact spectral boundary relation into an exact nonlinear determinant identity; numerical moment matching is not used.

### 3.2 From the path moments to the constrained determinant

*Proof of Theorem 3.1.* First remove $c$. If $J_c$ is the infinite symmetric tridiagonal matrix with diagonal $(c + x, c, c, \dots)$ and both off-diagonals equal to one, then $a_{m,k}(x, c) = (J_c^m)_{0k}$. Only a finite portion of this matrix is involved in each entry. Since $J_c = cI + J_0$,

$$a_{m,k}(x, c) = \sum_{j=0}^{m} \binom{m}{j} c^{m-j} a_{j,k}(x, 0).$$

Vandermonde's binomial convolution shows that the size-$n$ Hankel matrix at $c$ equals $B_n$ times the matrix at 0 times $B_n^{\mathsf{T}}$, where $(B_n)_{ij} = \binom{i}{j} c^{i-j}$ for $i \geq j$ and zero otherwise. The matrix $B_n$ is unitriangular, so

$$D_{k,n}(x, c) = D_{k,n}(x, 0). \tag{5}$$

Let $\mathcal{L}(f) = (f(J_0))_{00}$ and define monic polynomials

$$p_0(t) = 1, \qquad p_1(t) = t - x, \qquad p_{j+1}(t) = tp_j(t) - p_{j-1}(t) \quad (j \geq 1).$$

The recurrence gives $p_j(J_0)e_0 = e_j$, and therefore

$$\mathcal{L}(p_i p_j) = \delta_{ij}, \qquad \mathcal{L}(t^m p_k(t)) = a_{m,k}(x, 0).$$

For $k = 0$, the unit monic norms give $D_{0,n} = 1$ for every $n$. Then (1) is $1 - x^2 + (x^2 - 1) = 0$; this familiar case is included only for completeness. Henceforth assume $k \geq 1$. Thus $D_{k,n}$ is the Hankel determinant of the polynomial modification $p_k\mathcal{L}$ of an orthonormal functional with all monic norms equal to one. The classical Christoffel determinant identity, in the form of [4, Theorem 1], gives

$$D_{k,n}(x, 0) = (-1)^{nk} \frac{\det(p_{n+j}(\rho_i))_{1 \leq i \leq k,\, 0 \leq j < k}}{\Delta(\rho)}, \qquad \Delta(\rho) = \prod_{i<j} (\rho_j - \rho_i), \tag{6}$$

where $\rho_i$ are the roots of $p_k$, initially assumed distinct. Here is a formal-algebraic derivation, so no analytic measure is assumed. For $y = (y_1, \dots, y_n)$, expansion of the two determinants in the Vandermonde square gives the formal Andréief identity

$$D_{k,n} = \frac{1}{n!}\mathcal{L}^{\otimes n}\left[\Delta(y)^2 \prod_{j=1}^{n} p_k(y_j)\right].$$

The combined Vandermonde satisfies

$$\Delta(y, \rho) = (-1)^{nk}\Delta(y)\Delta(\rho)\prod_{j=1}^{n} p_k(y_j).$$

Replace its monomial columns by $p_0, p_1, \dots, p_{n+k-1}$, and replace those of $\Delta(y)$ by $p_0, \dots, p_{n-1}$; monicity leaves both determinants unchanged. Expand the combined determinant along its first $n$ rows. Orthogonality makes every term vanish unless the selected columns are exactly $p_0, \dots, p_{n-1}$. That surviving term has positive Laplace sign, and applying $\mathcal{L}^{\otimes n}$ contributes $n!$. The remaining minor is $(p_{n+j}(\rho_i))$. Substitution proves (6), with its stated sign and denominator. For $n = 0$ the identity reduces directly to monicity and both sides equal one.

Work now over generic $x$, choose $\rho_i = \lambda_i + \lambda_i^{-1}$, and put

$$A_i = \frac{\lambda_i - x}{\lambda_i - \lambda_i^{-1}}, \qquad B_i = \frac{x - \lambda_i^{-1}}{\lambda_i - \lambda_i^{-1}}.$$

The explicit solution of the three-term recurrence is

$$p_j(\rho_i) = A_i\lambda_i^j + B_i\lambda_i^{-j}.$$

The boundary condition $p_k(\rho_i) = 0$ says $B_i = -A_i\lambda_i^{2k}$ and also gives (3) with $K = k + 1$.

Define

$$R_i = \prod_{j\neq i} \frac{\lambda_i - \lambda_j^{-1}}{\lambda_i^{-1} - \lambda_j^{-1}}, \qquad w_i = \frac{A_i}{B_i}R_i.$$

Expanding the rows of the determinant in (6) into their two exponential terms gives

$$\det(p_{n+j}(\rho_i)) = \left(\prod_i B_i\right)\Delta(\lambda^{-1})\left(\prod_i \lambda_i\right)^{-n}\tau_n, \tag{7}$$

$$\tau_n = \sum_{S\subseteq\{1,\dots,k\}} \prod_{i\in S}(w_i\lambda_i^{2n}) \prod_{\substack{i<j\\ i,j\in S}} \left(\frac{\lambda_i - \lambda_j}{1 - \lambda_i\lambda_j}\right)^2.$$

Here $\Delta(\lambda^{-1}) = \prod_{i<j}(\lambda_j^{-1} - \lambda_i^{-1})$. To verify the expansion, first factor the term in which every row uses $B_i\lambda_i^{-n-j}$. Switching row $i$ alone contributes $(A_i/B_i)\lambda_i^{2n}R_i$. Switching both $i$ and $j$ corrects these individual contributions by the displayed squared factor. All Vandermonde factors are products over pairs, so these singleton and pair corrections account for every subset. The Cauchy determinant formula shows that this subset sum is exactly

$$\tau_n = \det\left(\delta_{ij} + \frac{(\lambda_i - \lambda_i^{-1})w_i\lambda_i^{2n}}{\lambda_i - \lambda_j^{-1}}\right)_{i,j=1}^{k}:$$

the principal minor indexed by $S$ is its corresponding summand. Consequently Lemma 3.3 applies.

Equations (6) and (7) give

$$D_{k,n}(x,0) = Cz^n\tau_n, \qquad C = \frac{(\prod_i B_i)\Delta(\lambda^{-1})}{\Delta(\rho)}, \qquad z = \frac{(-1)^k}{\prod_i \lambda_i}.$$

Every product in the centered identity (4) acquires the same factor $C^2z^{2m}$. Hence that identity holds with $\tau_j$ replaced by $D_{k,j}$. Set $m = n-k-1$; the three products then have the exact indices in (1).

Finally, the calculation is valid on a nonempty generic set of $x$. Indeed at $x=0$ the roots of $p_k = U_k(t/2)$ are distinct and lie in $(-2,2)$, so none equals 2 or $-2$; nearby generic nonzero $x$ avoids all the finitely many denominator conditions. For fixed $k,n$, the left side of (1) is a polynomial in $x$ and vanishes on this generic set. It is therefore the zero polynomial. This proves the identity also at $x=0,\pm1$, at root collisions, and at every zero of any Hankel determinant. Equation (5) restores arbitrary $c$. No division by an original Hankel determinant is used in the final extension. □

**Scope and attribution.** The theorem proves the entire stated recurrence, not merely the examples $k=1,2,3,4$, and not a determinant sequence with changed moments. The method passes through a new auxiliary spectral-closure question, but Christoffel modification, Cauchy minors, and the Hirota–Miwa equation are prior mathematics. The specific boundary-to-shift identification is the proposed contribution. A bounded search found no earlier proof of this exact Cigler recurrence; that does not certify universal historical priority or a historically unprecedented foundational principle.

## 4 A complete two-level tree classification and an excluded prior target

The original target considered here was Open Problem 7.1 of Coutinho–Guo–Schmeits [6]: whether infinitely many non-star finite trees admit Laplacian peak state transfer. A public construction of such trees was posted on 9 October 2026 and merged on 10 October in [7]. Therefore this existence statement is not counted as a newly solved open problem here. We nevertheless obtain a complete classification of every transferring vertex pair in the following natural two-parameter family. This is an auxiliary result, not a replacement for the original success criterion.

For integers $a \geq 1$ and $b \geq 2$, let $T(a,b)$ have a root $o$, children $c_1,\dots,c_b$, and $a$ private pendant leaves $\ell_{i,1},\dots,\ell_{i,a}$ attached to $c_i$. It is a non-star tree on $N = 1+b+ab$ vertices. For a real symmetric matrix $M = \sum_\lambda \lambda E_\lambda$ and distinct vertices $x,y$, nonzero peak transfer means

$$|(e^{itM})_{xy}| = \sum_\lambda |(E_\lambda)_{xy}| > 0.$$

Equivalently, all nonzero summands $e^{it\lambda}(E_\lambda)_{xy}$ have the same complex phase. For a connected graph Laplacian, $(E_0)_{xy} = 1/N > 0$, so that common phase must be 1.

**Theorem 4.1** (Classification of Laplacian peaks in two-level trees)**.** *Let $a \geq 1$, $b \geq 2$. Every distinct pair admitting Laplacian peak transfer in $T(a,b)$ contains the root $o$. Such a pair exists if and only if there are integers $u \geq 1$, $v \geq 2$ of opposite parity such that*

$$a = uv, \qquad b = uv - u + v.$$

*These parameters, when they exist, are unique. Set*

$$r = u(v-1)+1, \qquad s = v(u+1)+1, \qquad g = \gcd(r,s).$$

*Then the entire list of transferring pairs is as follows:*

1. *If $u$ is odd and $v$ is even, the root has peak transfer to every child, and to no leaf.*
2. *If $u$ is even and $v$ is odd, the root has peak transfer to every leaf, and to no child.*

*In either case the positive peak times are exactly the odd multiples of $\pi/g$. The peak amplitudes for the two cases are respectively*

$$B_L(o, c_i) = \frac{2(s-1)}{s(s-r)}, \qquad B_L(o, \ell_{i,j}) = \frac{2}{r(s-r)}.$$

*If the indicated parametrization with opposite parity does not exist, there is no Laplacian peak transfer between any distinct vertices. Equivalently, existence of a transferring pair is characterized by $a$ even, $b$ odd, and $(a-b)^2 + 4a$ being a perfect square.*

*Proof.* The normalized indicators of the root, all children, and all leaves span an invariant subspace for the Laplacian. Its matrix is

$$Q = \begin{pmatrix} b & -\sqrt{b} & 0 \\ -\sqrt{b} & a+1 & -\sqrt{a} \\ 0 & -\sqrt{a} & 1 \end{pmatrix}, \qquad p(z) := \det(zI - Q) = z(z^2 - (a+b+2)z + N).$$

Let $0 < r < s$ be the two positive roots. Since

$$(b-r)(b-s) = 1 - b < 0,$$

we have $r < b < s$. Also the quadratic has positive value $a(b-1)$ at 1 and negative value at $b > 1$, so its smaller root lies in $(1, b)$.

Since the root vector lies in this invariant subspace, it has zero projection onto its orthogonal complement. Dividing the quotient resolvent entries by the appropriate cell normalization factors gives for each individual child and leaf

$$((zI - L)^{-1})_{o,c_i} = -\frac{z-1}{p(z)}, \qquad ((zI - L)^{-1})_{o,\ell_{i,j}} = \frac{1}{p(z)}.$$

All three residues are nonzero. At the ordered eigenvalues $0, r, s$, the child coefficients have signs $(+, +, -)$, and the leaf coefficients have signs $(+, -, +)$. Thus the mutual support is exactly $\{0, r, s\}$.

If a peak occurs, the phases at $r$ and $s$ must both be real, so $tr, ts \in \pi\mathbb{Z}$. Hence $r/s$ is rational. As $r + s = a + b + 2$ is integral, both roots are rational; they are roots of a monic integer polynomial, so they are integers. Conversely, for integral $r, s$, write $r = gr_0$, $s = gs_0$ with $\gcd(r_0, s_0) = 1$. The integer combinations of $r_0, s_0$ show that $tg \in \pi\mathbb{Z}$ is necessary. The child sign pattern can occur exactly when $r_0$ is even and $s_0$ is odd; the leaf sign pattern requires $r_0$ odd and $s_0$ even. In each admissible case the allowed positive multiples of $\pi/g$ are exactly the odd ones.

It remains to express integrality and parity in the original graph parameters. The discriminant is

$$D = (a-b)^2 + 4a.$$

If it is a square $h^2$, then $h$ and $a - b$ have the same parity and $h > |a - b|$. Therefore

$$u = \frac{h + a - b}{2}, \qquad v = \frac{h - a + b}{2}$$

are positive integers with $uv = a$ and $b = uv - u + v$. The condition $b \geq 2$ excludes $v = 1$. Conversely these formulas give $D = (u + v)^2$ and the roots stated in the theorem. They also establish uniqueness. At least one of $r, s$ is odd: checking the four parities of $u, v$ gives

| $(u \bmod 2, v \bmod 2)$ | $r \bmod 2$ | $s \bmod 2$ |
|---|---|---|
| $(1, 0)$ | 0 | 1 |
| $(0, 1)$ | 1 | 0 |
| $(1, 1)$ | 1 | 1 |
| $(0, 0)$ | 1 | 1. |

Thus $g$ is odd and the required root parities are exactly the two opposite-parity cases in the theorem. Since the sum of all cross projector coefficients is zero, the peak amplitude is twice the absolute value of the unique negative coefficient. The residues give $-(s-1)/(s(s-r))$ for a child and $-1/(r(s-r))$ for a leaf, proving the stated amplitudes.

Finally consider two distinct nonroot vertices. Besides the radial subspace above, the Laplacian has a branch-contrast subspace. Choose $t = (t_1, \ldots, t_b)$ with $\sum_i t_i = 0$. Vectors with child values $xt_i$ and all leaf values $yt_i/\sqrt{a}$ in branch $i$ reduce the Laplacian to

$$R = \begin{pmatrix} a+1 & -\sqrt{a} \\ -\sqrt{a} & 1 \end{pmatrix}.$$

Its eigenvalues $\rho_- < \rho_+$ are the two roots of $z^2 - (a+2)z + 1$. They are irrational, because

$$(a+1)^2 < a^2 + 4a < (a+2)^2 \qquad (a \geq 1).$$

Both coordinates of either normalized eigenvector of $R$ are nonzero. The remaining leaf-contrast subspaces consist of vectors summing to zero within a private leaf set and have eigenvalue 1. The dimensions $3 + 2(b-1) + b(a-1) = N$ account for the full vertex space. For nonroot vertices in branches $i, j$, the cross entry of the $\rho_\pm$ projector is the product of these nonzero coordinates (with a factor $1/\sqrt{a}$ per leaf), multiplied by $\delta_{ij} - 1/b$. This factor never vanishes for $b \geq 2$. There is no eigenvalue collision with the radial subspace: subtracting the two quadratics would force a common root to equal $a + 1$, whereas $R$'s characteristic polynomial at $a + 1$ equals $-a$. Nor is either root equal to the leaf-difference eigenvalue 1. Consequently both $\rho_-, \rho_+$ lie in the mutual support of every nonroot pair. The zero eigenvalue also lies in that support. Peak alignment would imply $\rho_+/\rho_- \in \mathbb{Q}$, which, combined with $\rho_+ + \rho_- = a + 2$, would make both roots rational. This contradiction excludes every nonroot pair and completes the classification. □

Taking $u = 1$, $v = 2k$ gives the particularly small infinite family

$$a = 2k, \qquad b = 4k - 1, \qquad N = 2k(4k+1), \qquad k \geq 1,$$

with root-to-child peak transfer at time $\pi$. Its first orders are $10, 36, 78, 136, \ldots$. Taking $u = 2$, $v = 3$ gives a 50-vertex tree with root-to-leaf peaks at time $\pi/5$. The public prior family for the original existence question is instead $a = t(t+1)$, $b = a + 1$, for positive odd $t$, with roots $t^2 + 1$ and $(t+1)^2 + 1$. That prior construction already settles the unchanged existence question. The classification above supplies more detail within a restricted family, but does not constitute a first resolution of Open Problem 7.1 or a classification of all trees.

The proof uses equitable subspaces, spectral projectors, and elementary integer factorization. These are classical tools. A historically new foundational method is not asserted.

# A Least dimension: an unsuccessful projection route

Clarke's Question 6.8 [8] asks for the least dimension of an order-two counterexample to polynomiality implying effectiveness. We do *not* resolve it here. Combining the source's six-dimensional example with the preceding T83 internal proof [10] gives only $4 \le d_{\min} \le 6$. The obstruction below excludes a particular route to dimension five; it is not counted as a new named-problem solution.

Let $P_0 \subset \mathbb{R}^6$ be the convex hull of

$$\begin{array}{llll} (1,0,0,0,0,0), & (0,0,1,0,0,0), & (0,1,0,1,1,0), & (0,0,1,1,1,0), \\ (0,0,0,1,0,0), & (0,0,0,0,0,1), & (1,1,0,0,1,0), & (1,1,0,0,0,1). \end{array}$$

This is Clarke's Example 5.6, with involution $\sigma(u,v) = (v,u)$. For $a, b \in \mathbb{Z}$, define the surjective lattice quotient

$$T_{a,b} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & a \\ 0 & 1 & 0 & 0 & 0 & b \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & -a \\ 0 & 0 & 0 & 0 & 1 & -b \end{pmatrix}, \qquad Q_{a,b} = T_{a,b}(P_0).$$

Its kernel is generated by the primitive anti-invariant vector $(a, b, 1, -a, -b, -1)$. The induced involution is

$$(y_1, y_2, y_3, y_4, y_5) \longmapsto (y_4 + a y_3, y_5 + b y_3, y_3, y_1 - a y_3, y_2 - b y_3).$$

Consequently $Q_{a,b}$ is five-dimensional, and the homogeneous plus/minus eigenspace dimensions are $(p, q) = (4, 2)$.

**Projection obstruction.** For every $a, b \in \mathbb{Z}$, the equivariant numerator of $Q_{a,b}$ is not polynomial.

*Proof.* The fixed lattice has coordinates $(y_4, y_5, y_3)$. Averaging the vertices shows that twice the fixed polytope $R_{a,b} = Q_{a,b}^{\sigma}$ is

$$2R_{a,b} = \operatorname{conv}\{(-a, -b, 2), (1, 0, 0), (1 - a, 1 - b, 2), (1, 2, 0)\}.$$

Changing $(a, b)$ to $(a + 2r, b + 2s)$ gives the lattice shear $(x, y, z) \mapsto (x - rz, y - sz, z)$. Moreover the affine lattice involution $(x, y, z) \mapsto (1 - x - z, 1 - y - z, z)$ sends $R_{0,0}$ to $R_{1,1}$ and $R_{0,1}$ to $R_{1,0}$. Thus there are only two counting cases.

Here is an explicit finite certificate, independent of any hull software. Put

$$A_{k,z} = \max\{0, \lfloor k/2 \rfloor - \lceil (k - z)/2 \rceil + 1\}.$$

The four supporting inequalities of $R_{0,0}$ give, in $kR_{0,0}$,

$$0 \le z \le k, \quad (k - z)/2 \le x \le k/2, \quad x - (k - z)/2 \le y \le x + (k - z)/2.$$

For $kR_{0,1}$, the $x, z$ bounds are identical and the $y$ bounds become $x - k/2 \le y \le x - z + k/2$. Therefore their lattice counts are exactly

$$N_{0,0}(k) = \sum_{z=0}^{k} A_{k,z}\,(2\lfloor (k - z)/2 \rfloor + 1),$$

$$N_{0,1}(k) = \sum_{z=0}^{k} A_{k,z}\,(2\lfloor k/2 \rfloor - z + 1).$$

For $k = 0, \dots, 7$ these sums give

| $k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $N_{0,0}(k)$ | 1 | 1 | 6 | 6 | 19 | 19 | 44 | 44 |
| $N_{0,1}(k)$ | 1 | 0 | 7 | 3 | 22 | 13 | 50 | 34 |

Each $R_{a,b}$ has dimension three and denominator at most two, so its even and odd Ehrhart constituents have degree at most three. Four values of each parity determine them, giving

| | $a \equiv b \pmod 2$ | $a \not\equiv b \pmod 2$ |
|---|---|---|
| $N_{\mathrm{even}}(k)$ | $\dfrac{(k+2)(k^2+4k+6)}{12}$ | $\dfrac{(k+2)(k+4)(2k+3)}{24}$ |
| $N_{\mathrm{odd}}(k)$ | $\dfrac{(k+1)(k^2+2k+3)}{12}$ | $\dfrac{(k-1)(k+1)(2k+3)}{24}$ |
| $N_{\mathrm{even}} - N_{\mathrm{odd}}$ | $\dfrac{k^2+3k+3}{4}$ | $\dfrac{(2k+3)^2}{8}$. |

In both cases the difference has degree two, hence the fixed Ehrhart series has a pole of order three at $t = -1$. The character formula

$$\varphi_{Q_{a,b}}(t)(\sigma) = (1-t)^4(1+t)^2 \sum_{k \geq 0} N_{a,b}(k) t^k$$

cancels only two orders. This proves the claimed nonpolynomiality for all integer parameters, not just those in a bounded search. □

**A rejected four-dimensional candidate.** Under $(x_1, x_2, y_1, y_2) \mapsto (y_1, y_2, x_1, x_2)$, consider

$$P = \mathrm{conv}\{(1,0,0,0), (0,0,1,0), (1,1,0,1), (0,1,1,1), (2,1,1,0), (1,0,2,1)\}.$$

Its fixed triangle is $\mathrm{conv}\{(1/2, 0), (1/2, 1), (3/2, 1/2)\}$, with constituents $N_{\mathrm{even}}(k) = k^2/2 + k + 1$ and $N_{\mathrm{odd}}(k) = (k^2 - 1)/2$. Exact counts $1, 6, 23, 68, 165$ at dilations $0, \dots, 4$ give $h^*_P = 1 + t + 3t^2 + 3t^3$, and consequently

$$\varphi_P = (1 + 3t^2 + 2t^3)\chi_0 + (t + t^3)\chi_1.$$

It is effective and therefore fails as a counterexample. The accompanying exact-arithmetic certificate supplies its supporting facets and verifies all the displayed counts. Neither this rejection nor the projection obstruction supplies the missing four- and five-dimensional classification.

# B A sparse-PT construction that cannot solve the odd-period problem

Faust–Friedman–O'Malley–Ramos–Sharma, Conjecture 2 in *Algebraic Properties of the Ideal of Spectral Invariants for the Discrete Laplacian* [9], asks for a nonzero reversal-conjugate $q$-periodic potential Floquet isospectral to zero for every $q > 3$. Their source proves every even period. The uniform odd-period assertion is not proved here. The following proposition rules out one proposed uniform construction; it is not a solution of that conjecture.

**Proposition B.1** (Five-site reflected obstruction)**.** *Let $r \geq 1$ and $q = 2r + 5$. For real $a, b, c, d$, put*

$$V = (a + ib, c + id, \underbrace{0, \dots, 0}_{r}, -2(a + c), \underbrace{0, \dots, 0}_{r}, c - id, a - ib).$$

*If the nearest-neighbor discrete Schrödinger operator with potential $V$ and off-diagonal weights one is Floquet isospectral to zero, then $a = b = c = d = 0$.*

*Proof.* Write

$$M(z) = \begin{pmatrix} x - z & -1 \\ 1 & 0 \end{pmatrix}, \quad A = M(0), \quad C = M(a + ib), \quad B = M(c + id), \quad h = -2(a + c),$$

and put $D = BC\overline{C}\,\overline{B}$, where conjugation fixes $x$. By cyclicity of trace, the transfer discriminant of $V$ is $\operatorname{tr}(A^r M(h) A^r D)$; that of zero is $\operatorname{tr}(A^{2r+5})$. Set $u = U_r(x/2)$ and $v = U_{r-1}(x/2)$ for the Chebyshev polynomials of second kind. Then

$$A^r = \begin{pmatrix} u & -v \\ v & u - xv \end{pmatrix}.$$

The discriminant difference has the form

$$F(x)u^2 + G(x)uv + H(x)v^2, \qquad H(x) = (-3a^2 + b^2)x + 2a(a^2 + b^2).$$

The formula for $H$ follows by substituting $u = 0, v = 1$ in the matrix expression; it is a polynomial identity independent of $r$. Consecutive Chebyshev polynomials are coprime. Consequently a vanishing discriminant difference implies $u \mid H$. If $r = 1$, then $u = x$, so the constant coefficient of $H$ vanishes. If $r \geq 2$, then $\deg u \geq 2 > \deg H$, so $H = 0$. In either case $a(a^2 + b^2) = 0$, and reality gives $a = 0$.

The coefficients of $x^{q-2}$ and $x^{q-3}$ in the discriminant difference are respectively

$$-3a^2 - 4ac + b^2 - 3c^2 + d^2, \qquad 2(a^3 + 4a^2c + ab^2 + 4ac^2 + c^3 + cd^2).$$

These follow directly from the first three trace identities, with $\sum_j V_j = 0$. Substituting $a = 0$ into the second coefficient gives $c(c^2 + d^2) = 0$, hence $c = 0$. The first coefficient is then $b^2 + d^2$, so $b = d = 0$. □

The central value was imposed by the trace, so the proposition exhausts this five-site reflected pattern. Period five corresponds to $r = 0$; then $u = 1$ gives no restriction, consistently with the nonzero solutions already certified in the source. Exact symbolic calculations verify the transfer identity and coefficient formulas for $q = 7, 9, 11, 13, 15, 17$; the divisibility argument proves the assertion for every $r \geq 1$.

This removes a candidate construction, but does not imply nonexistence of less sparse PT potentials. The original all-odd-period conjecture and the unrestricted period-lattice classification remain unresolved in this attempt. Transfer matrices, Chebyshev identities and polynomial coprimality are classical ingredients; no historically new foundational principle is claimed.

## C Verification, attribution, and the success threshold

The accompanying archive supplies seven executable Python verification programs and their recorded JSON outputs. Each uses exact integer, rational, or symbolic arithmetic. Finite tests check implementation and formula consistency; the preceding proofs establish the infinite quantifiers. The distinction is especially important for the Hankel recurrence, whose early zero blocks can make short tests vacuous.

| Component | Executed checks |
|---|---|
| Stern families $0, 2, 3$ | Eight exact Bernstein identities; 177 independent Stern-recursion comparisons, covering all three families for $2 \le n \le 60$. |
| Fourth Stern family | Sixty exact factorization checks; twelve exact Sturm and multiplicity checks. The proof separately covers every $n$. |
| Cigler recurrence | 108 parameter runs comparing direct Hankel determinants with a fixed-size reduction; 24 long-range recurrence checks for $1 \le k \le 12$, $x = 2, 3$, reaching $n = 512$. |
| Cauchy identity | Exact rational Cauchy-minor expansions and nontrivial three-direction identities for one through six spectral pairs. |
| Tree classification | 870 parameter cases and seven full Laplacian matrices of trees with at most 50 vertices. Individual vertex projectors, not only quotient states, are checked. |
| PT obstruction | Exact symbolic transfer discriminants and coefficient identities for periods $7, 9, 11, 13, 15, 17$. The Chebyshev divisibility proof covers all odd periods at least seven in the stated pattern. |
| Ehrhart attempts | Exact supporting-facet and lattice-point calculations for the four-dimensional candidate and the four parity classes of the projected fixed polytopes. |

Independent internal reviews rederived the Stern identities and all positivity certificates, and audited the fourth-family root counts. For the Hankel theorem, they independently checked the Cauchy rank-two identity, reciprocal-root shifts, Vandermonde normalization, and the extension to exceptional parameters. An alternative Casoratian–Plücker derivation was also reconstructed during the audit. These reviews are not external peer review or formal proof-assistant certification.

The original named targets that receive complete, unambiguous arguments in this manuscript are Ulas Conjecture 4.3(1) and Cigler Conjecture 4.1. The fourth Stern family is fully classified, but the multiplicity ambiguity of its source statement prevents treating it as a separate unqualified historical closure. The already preceded tree existence question, the two unresolved main questions, and the initial Motzkin shifted-period attempt are excluded from this count.

The priority search inspected the original papers, author-hosted material, follow-up literature, and targeted public proof records through 26 October 2026. No earlier identical complete proof of the two principal applications was located. This is the outcome of a bounded search, not a theorem about all prior literature. The general methods have classical antecedents, explicitly credited in the body and bibliography. Thus the full demand for five historically unprecedented-method solutions is not reported as achieved.

The standalone TEX embeds its bibliography and uses no external figures or bibliography database. It compiles with two passes of `pdflatex`. The archive includes dependency information, run instructions, and SHA-256 checksums. Recorded calculation outputs are provided to make the finite checks inspectable and reproducible.

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